



Specification

Fire Sprinkler Protection & Dry Fire Services

SPECIFICATION
F25-0437

ISSUE
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ARCHITECT
Archipelago

Project Name: Skygate Play - Brisbane Airport
At The Circuit, Brisbane Airport



Electrical



Mechanical



Hydraulic



Energy

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QA System

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A	17 December 2025	Issue for D&C Pricing	DE
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1. PRELIMINARIES

1.1 Conditions of Contract

This section of the specification forms part of the contract documents. Relevant sections of the main contract documents shall be read in conjunction with this part of the specification.

2. SCOPE

2.1 General

The work carried out shall comprise the manufacture, supply, delivery, installation, testing, commissioning and maintenance for the specified period of the Fire Protection System, as described in detail in the following Clauses and shown on the Drawings. This performance document outlines the Fire Protection and Dry Fire Services for the project.

It is based on the drawings and related documents produced as part of the D&C design phase. As such it reflects the "D&C Pricing" stage of documentation. The Fire Services sub-contractor shall make all allowances necessary to price the project, based on a Design & Construct methodology and style of project delivery.

This document and any associated documents have been produced on the premise that they provide descriptions and information of sufficient detail to enable a full evaluation of the complete project to be gained by:

- Using the descriptions and details provided.
- Relying on the tender's expertise to make the appropriate allowances by extrapolations, inference, deduction, assumption and the like to ensure a complete and operational fire sprinkler services system is provided that meets and satisfies the clients expectations and operational requirements.
- Where items are not specifically shown on the drawings or included within this document but are required for the type of facility proposed, taking the quality and extent of the provision from the standards set by the information provided in the documentation and including for all such requirements.

This document is NOT intended to be an exhaustive brief for the project. It is intended to:

- Set a minimum standard for the quality both in materials and workmanship for all fire sprinkler services works and systems.
- Be a basis for the production of documentation to be used to lodge for Building Approval (complete with Form 15 design certification), QFES approvals and For Construction Documentation.

The fire services drawings and specification indicate the concept design requirements for this project and set about the scope to allow competitive tenders to be called.

The contractor shall be responsible for the completion of design and detailed services co-ordination and all design drawings shall be submitted to the project manager by the contractor for final approval, prior to lodgement with all required authorities.

All work shall be carried out in accordance with the Specification, Drawings and such details and instructions as are issued by the Superintendent and/or Consulting Engineers during the progress of the work and to the entire satisfaction of the Superintendent.

All work shall be carried out strictly in accordance with manufacturer's recommendations and industry best practice. The equipment installed shall be handed over to the client in a fully operational state as intended for the equipment function and use.

This Specification shall be read in conjunction with the Drawings, all of which are intended to be mutually explanatory. All work called for by the one even if not by the other, shall be fully executed. The Drawings and Specification are intended to include everything necessary for the proper completion and setting into effective operation of the installation shown. Minor items as used in normally accepted trade practice with installations of this type and which are not specifically mentioned, shall be included in the Tender and installation.

This Specification relates to workmanship and materials used in the work and shall be applied to equipment supplied or installed by any trade, company, or assisting firm engaged to supply or install any materials and/or equipment.

2.2 Fire Engineering Report

The project is subject to a Fire Engineered solution for a number of issues. The Fire Engineering Report shall be perused to ensure compliance.

Provide and install a fire protection installation in compliance with the Building Code of Australia and the Fire Engineering Report.

2.3 Fire Sprinkler Options Note

The project has identified a number of options to the sprinkler water supply. The base option for pricing shall be Option 1 as nominated on the design documents. Options 2 and 3 shall be priced as alternatives.

2.3.1 Option 1

The baseline for pricing shall be as shown on the drawings, establishing water storage tank and pump house with dual diesel sprinkler pumps adjacent to the Snow World component of the proposed Area 51 tenancy (co-located with hydrant pump). Sprinkler valve sets proposed in pump room.

2.3.2 Option 2

An opportunity has been identified to utilise the existing fire sprinkler infrastructure of the adjacent Skygate Home & Life site, located on the far end of that project. It will require cutting into an existing pipe just where it comes up from inground, and then continuing this to above roof, travel the length of that building to drop down on opposite side and travel inground under road to the corner of the Game Over tenancy of new Skygate project, where the valve sets would be established.

The FDCIE's for the Skygate Home & Life and the new Skygate shall be networked together to allow transfer of all events as deemed necessary. The FDCIE at Skygate Home & Life shall be upgraded to a Notifier Networkable type.

2.3.3 Option 3

The final pricing option is similar to option 1, except that all infrastructure is now proposed on the opposite side of the (service) road, with the view that this infrastructure can be used for another future development in that area, and with the purpose of freeing up the front of this project for possible future tenancy frontage. This option will require inground piping to a header with valve sets on the corner of the new Skygate building, and a stub for future connection to future development.

2.4 Extent of Work

The work includes but is not limited to the following main elements:

- Design and Construction of the Fire Services as detailed in this brief and the drawings. All works to be in accordance with NCC Statutory requirements, Brisbane Airport Corporation, Fire Engineering Report and relevant Australian Standard requirements.
- The successful Fire Services Contractor shall be responsible for a fully designed, engineered and certified system.
- Greenstar is proposed for this building, as such the design and construction of the Fire services shall align with the selected points table as issued by the Greenstar Professional.

Fire Sprinkler Services

- Automatic sprinkler system c/w storage tank and pumps
- Shop drawings, detailed coordination and pipe sizing calculations
- Final fire pump duty calculations
- Water storage calculations
- Cable and pipe supports, fixings, fittings, and associated equipment
- Trenching and backfilling
- Transport to site, positioning and hoisting of equipment
- Provision of workshop drawings
- Penetrations through fire and smoke walls and sealing upon completion

- Testing and commissioning
- Maintenance and defects liability (12 months)
- Provision of operation and maintenance manuals and “as installed” drawings.
- Provision of Baseline Data in accordance with the requirements of AS 1851 and 1670.1.
- Provision of Certification of Compliance.
- Instruction of the client in the operation and use of the fire protection systems.
- Any additional work necessary to provide a complete, operative installation in accordance with performance requirements and miscellaneous works, as specified and shown on the Drawings.

Dry Fire Services

- Analogue addressable system throughout
- Connection to fire brigade or monitoring service provider via ROMTECH RM3118 (or as required by the fire brigade to communicate with the headend) Dual SIM ASE however confirm with BAC, pay all costs for the installation of this by a licensed contractor
- Control and indicating Equipment (CIE’s)
- Replace existing Control and Indicating Equipment in Skygate Home & Life
- Provide network cable between existing and new FDCIE’s
- Fire fan control panel
- Locality panels
- Smoke detectors
- Heat Detectors
- Multi sensor detectors
- Manual call points
- Trenching, backfilling, conduits and pits
- Audible and visual alarms
- Interfaces with sprinkler, mechanical, access control, hydraulic services etc
- Air conditioning shutdown requirements
- Alarm signal to open vehicle gates in the event of a fire alarm
- Monitoring of all sprinkler equipment, pumps, flow devices etc
- Zoning design
- Programming
- Building occupant warning system
- Fire extinguishers
- Peripheral devices as required
- Surge and lightning protection
- Fire stopping of penetrations
- Authority fees
- Testing and commissioning
- Operation manuals and as built drawings
- Training of personnel

The installation shall be carried out by a QBCC licensed fire system installation company.

2.5 Design Requirements

The following requirements apply to all BAC controlled buildings within Brisbane Airport excluding the Domestic Terminal Building (DTB) and International Terminal Building (ITB) for which separate design guidelines exist.

- Where a building requires a fire system of any description as a requirement of the National Construction Code of Australia, Australian Standards, these fire systems shall not be removed by means of a fire engineered solution.
- Where required, all new fire systems are to comply with FM Global fire system design requirements.
- All fire system equipment such as valves, sprinklers and the like shall be to Australian Standards and listed for use in a fire system except where watermark approval for equipment connected to drinkable water supplies is required.
- External fire alarm panels and associated building occupant warning systems are to be housed in IP65 rated stainless steel cabinets.

2.5.1 Design Process

The following process shall be followed to ensure the design meets the requirements of BAC, ABC and other relevant stakeholders. Refer also to referenced documents for further details of the fire system procedures.

- Designer to request "As Constructed" information from BAC regarding the existing building services in the proposed area of work.
- Designer to request current version of the FSAR, FER and MIU documentation from BAC for the proposed area of work.
- Designer to undertake a site survey of the area where the proposed works are to occur to check the validity of the "As Constructed" documentation.
- The designer to ensure they have the latest revision of the fire matrix for the building that the fire design is being undertaken.
- Designer and/or installer to submit required documentation for BAC approval in accordance with the Design Deliverables Section of this Brief.

2.5.2 Drawings & Design Responsibility

The Fire services drawings supplied are diagrammatic and approximate only.

These drawings, together with this performance document and all associated consultant's drawings and briefs are intended to be mutually explanatory and to show the scope of the work required for this D&C Pricing Stage. All work set forth by one, if not by the other, shall be fully executed.

The work covered within the drawings and documents produced by the fire sprinkler services sub-contractor shall comprise the finalisation of design, co-ordination, documentation, safety in design and construction, supply, installation, testing, commissioning, quality assurance, certification, and maintenance during the defects liability period for the provision of the fire sprinkler services systems as indicated within this document and the following drawings to serve the development.

The fire services sub-contractor shall be responsible for the detailed design activities listed below in addition to those activities normally undertaken during construction and shall achieve occupational approval for the overall facility.

The detailed design shall be fully co-ordinated and compatible with the remainder of the project design.

Design responsibilities include:

- Production of client approval drawings.
- Production of co-ordination drawings.
- Production of and submission of drawings for authority approval.
- Production of For Construction drawings.
- Production of As-built drawings.
- Undertake detailed spatial co-ordination of the services with respect to all other works and trades including the detailing of interfaces. The contractor is responsible for ensuring all interfaces are covered and that there no scope gaps.
- Perform & confirm final calculations for all sprinkler protection services requirements, including pipe sizing calculations using approved software, pump duties, pipe sizes & velocities, pressure losses, pump selections, etc.
- Design and co-ordinate the supports for the services and equipment.
- Design and select fire stopping for all services penetrations.
- Size and location of penetrations.
- Coordination of interfaces for all trades. Whether or not explicitly identified in the tender documentation, the contractor shall establish all the required interfaces between each service and coordinate the provision of such interfaces. This shall be considered when preparing tender price and undertaking final design.
- Co-ordination of installation with other trades.
- Co-ordination of the construction of the installation.
- Accommodate thermal expansion and construction joint movement in all services taking into account final installation details and consistency with the specified requirements.
- Final selection of all valves, pipework, ductwork, plant and equipment.
- The preparation of fire sprinkler services drawings sufficient to allow construction, including submissions where required for approvals and authority permits.
- Allow for detailed co-ordination in specific locations eg. Plant rooms, risers, corridors, ceiling & floor voids, etc.
- Technical co-ordination of all the elements of the fire sprinkler services works and any suppliers.
- Where the contractor wishes to use alternative methods or equipment, fully evaluate and submit a detailed report on any proposals for review by the Principles Representative.

2.5.3 Design Life

Element	Minimum Design Life
Fire detection systems	15 years
Fire alarms including EWIS and PA systems	15 years
Fire Sprinkler systems	25 years

2.5.4 Design Criteria

When the fire services design is being undertaken there should be consideration given to a number of other factors that may be outside of the design guidelines, technical specifications and referenced standards that should be considered. Such considerations include the following (not exhaustive):

- Use and function of the location being designed
 - When the system is located in a terminal is it a public area or is it an area of storage?
 - Does the area have special building services located in the space such as baggage conveyors etc?
- Fire engineering requirements
 - The fire engineering aspects have a large impact on the wet and dry fire systems along with evacuation systems.
 - The zoning can be impacted that requires upgrades to a number of smaller systems to achieve compliance.
- Site infrastructure capacity
 - Does the water supply to the building have enough capacity for the control of the fire?
 - Does the batteries in the EWIS panel have enough capacity for door holders or other equipment that is to be relied upon in a mains power supply failure?
- Whole of life costs
 - Is the system right for the area of use or could a more maintenance friendly solution be used even if the initial capital cost may increase?
- Ecologically sustainable design (ESD)
 - Is test water being recovered or recycled and where can that water be used?
 - How will unwanted fire system components be removed and disposed of in the future?
- Maintenance
 - Have the needs of the maintenance personnel been considered in the design?
 - For example, where are isolation valves and panels located?
 - Can detectors be accessed for maintenance without EWP?
- Operations
 - Does the location of equipment impact the operation of the terminal or building?
 - For example, is there sufficient clearance above walkways, driveways, baggage conveyors, etc?
- Value
 - Will the design give the best value to BAC?

2.5.5 Change Management

Any changes to existing fire services software shall be made by the relevant Contractor based on BAC change management process as described in PR-5505 Fire System Procedures.

2.5.6 Design Deliverables

Project Phase	Milestone	Description of Documentation Required
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Schematic	15%	Identify whether a fire system is required and if there is adequate existing infrastructure available. Determine appropriate hazard classifications (eg Ordinary Hazard 1) referencing applicable standards, FSARs and FERs. Refer to fire matrix to provide preliminary identification of fire detection and ancillary device numbering information. Refer to fire matrix to provide preliminary identification of mechanical / security services and fire system interfacing. Identify fire system zoning. Identify location of major plant and equipment. Identify location of main service routes (pipes, cable supports, etc). Identify interfaces to other systems. Identify any other requirements imposed by existing FSARs and FERs.
Developed Design	40%	Updated Schematic documentation. Fire engineering brief (FEB) where applicable - If not applicable provide a statement confirming as such. Preliminary design specifications. Preliminary locations of minor plant and equipment. Confirm fire detection ancillary device numbering including network, loop and device numbering along with mechanical / security interfaces. Confirm hydraulic calculations to meet referenced standards and FM Global requirements. Confirm system capacity exists or needs to be increased to suit the proposed works including but not limited to: - Confirm fire and/or smoke compartmentation

		- Confirm fire systems zoning - Confirm location of major plant and equipment - Confirm location of main service routes (pipes, cable supports, etc) - Confirm interfaces to other systems - Confirm any other requirements imposed by existing FSARs and FERs - Fire alarm network capacity - EWIS/PA system capacities (including amplifiers) - Battery capacities - Provision of preliminary fire matrix information Confirm interfaces to other systems such as: - Mechanical systems (smoke management, shutdown, etc) - Security systems (door access control release, lifts, etc) - Interfaces between base building and tenancy works By the use of the fire matrix and other fire and other services Design Guides. Update of fire matrix prior to Detail Design regarding all fire systems and other building services interfaces. Cause and effect matrix where applicable (mandatory for DTB and ITB). EWIS / PA cascading matrix where applicable (mandatory for DTB and ITB).
Detailed Design	90%	Updated Developed Design documentation. Fire engineering report (FER) where applicable - If not applicable provide a statement confirming as such. Design documentation and certification suitable for submission to Airport Building Controller. Fire evacuation plan changes and/or updates. Complete update of fire matrix prior to construction issue regarding all fire systems and other building services interfaces. Provide fire matrix prior to construction regarding all fire systems and other building services interfaces.

		Provide cause and effect matrix where applicable (mandatory for DTB and ITB). Provide EWIS/PA cascading matrix where applicable (mandatory for DTB and ITB). Provide screenshots of updates to Fire Graphics System (FireNet Digifire at ITB, Simplex Truesite at DTB).
Construction	100%	Updated Detailed Design documentation addressing all BAC and Airport Building Controller comments.
Pre-Construction (Workshop drawing phase)		Testing and commissioning methodology statement. Fire impairment methodology. Fire detection equipment numbering.
Post-Construction		As-built drawings, schematics and manuals. As-built fire detection numbering. Cause and effect matrix where applicable (mandatory for DTB & ITB). EWIS/PA cascading matrix where applicable (mandatory for DTB and ITB). Installation certification. Testing and commissioning records. Where applicable, documents shall be updates of existing as-built records. Post-construction documentation that is separate from or does not update existing records will not be accepted. All other documentation required by PR-5505.

2.5.7 Design Specification Requirements

The design specification is to include the following as a minimum:

- Scope of the works being designed
- Interfaces with building and BAC infrastructure
- Interfaces required between services
- Fire Matrix overview
- Materials
- List of fire services drawings
- Scope of changes to Fire Graphics packages - FireNet Digifire at ITB, Simplex Truesite

- at DTB
- Returnable schedules

2.6 Work in Other Trades

Related work in connection with the Fire Protection Services (performed by other trades) includes the following:

2.6.1 By Main Contractor

- Provision of access to concealed detectors.
- Formed and trimmed holes in ceilings, walls and floors.
- Water supply into sprinkler water storage tank.
- Drain point for test drains in the control valve locations and the sprinkler system extremity test points
- Drain points at tank overflow and in sprinkler pump room
- Hydrant and hose reel systems
- Enclosing of pipework risers.
- Co-ordination with fire protection services to allow installation of pipework and wiring concealed in ceilings, floors, walls and partitions.
- Fire zone fire door sets complete with automatic closers, smoke seals and fire rated door frames.
- Fire roller shutter and electromagnetic link in clutch mechanism.
- Fire protection services equipment cupboards with the doors suitably labelled as to their contents (in 50 mm high lettering which contrasts to the door finish).

2.6.2 By Electrical Services

- 240V AC supply to equipment
- Power supply (lockable isolator) adjacent to the sprinkler control valve set.
- Emergency lighting at Main CIE and Fire Sprinkler control valve sets.
- Provision of interfaces.

2.6.3 By Mechanical Services

- Co-ordination with the Mechanical Services to allow installation of detectors associated with mechanical plant and cables to air conditioning fire control relays.
- Provision of fire control relays in Mechanical Switchboards

3. GENERAL REQUIREMENTS

3.1 General

This section outlines general requirements which are applicable to all sections of this specification and the works.

3.2 Standards and Reference Documents

The fire services design shall be based on the scope noted above together with the requirement in all required standards, guidelines and reference documents. The documents listed below are the minimum requirements for the design:

Reference	Title
AAG-615	Hydraulic Connections
ADG-001	Introduction and Overview
ADG-010	Design Principles
ADG-020	Sustainable Design
ADG-050	Project Completion Documents
ADG-051	Commissioning
ADG-525	Skygate Tenancy Design
ADG-601	Electrical General
ADG-610	Fire
ADG-615	Hydraulic
ADG-625	Building Management System
ADG-620	Mechanical
ADG-903	BIM and CAD Standards
ADG-616	Fuel
DG-1031	Adhesives, sealants and Fasteners
DG-1045	Waterproofing
DG-1147	Signage
TS-5505	Fire Systems Procedures
	Fire engineering reports (FER) and fire safety assessment reports (FSAR) for each building as applicable - Current issue to be requested at commencement of design
	Maintenance in use (MIU) plan for each building as applicable - Current issue to be requested at commencement of design

3.3 Industry Standards, Regulations and Guidelines

Reference	Title
AS/NZS 1221	Fire Hose Reels
AS 1345	Identification of the contents of pipes, conduits and ducts
AS 1530	Methods for fire tests on building materials, components and structures
AS/NZS 1668	The use of ventilation and air conditioning in buildings
AS 1670	Fire detection, warning, control and intercom systems - System design, installation and commissioning
AS/NZS 1841	Portable fire extinguishers
AS/NZS 1850	Portable fire extinguishers - Classification, rating and performance testing
AS 1851	Routine service of fire protection systems and equipment
AS/NZS 2053	Conduits and fittings for electrical installations
AS 2118	Automatic fire sprinkler systems
AS 2419	Fire hydrant installation
AS 2441	Installation of fire hose reels
AS 2444	Portable fire extinguishers and fire blankets - Selection and location
AS/NZS 3000	Electrical installations (known as the Australian/New Zealand Wiring Rules)
AS/NZS 3008	Electrical installations - Selection of cables - Cables for alternating voltages up to and including 0.6/1 kV
AS 3745	Planning for emergencies in facilities
AS 4072	Components for the protection of opening in fire-resistant separating elements - Service penetrations and control joints.
AS 4428	Fire detection, warning, control and intercom systems
AS 7240	Fire detection and alarm systems
AS 60529	Degrees of protection provided by enclosures (IP Code)
AS 60849	Sound systems for emergency purposes
	Factory Mutual (FM) Global loss prevention data sheets

3.4 Quality of Materials and Workmanship

All materials shall be new, of the best quality and of the class most suitable for the purpose specified.

All equipment shall have a verified history of successful use in the commercial environment. No prototype equipment will be accepted.

All work on the site shall be supervised by a competent tradesman supervisor who shall be present on site when requested by the Superintendent, for all phases of the project. A representative shall be nominated, who shall be present and responsible for the day to day running of the project.

The fire services installation shall be executed by licensed, experienced tradesmen, and shall be completed in a neat and first-class tradesman-like manner to the satisfaction of the Superintendent.

3.5 Uniformity

All fittings and equipment of the same type shall be of the same manufacture and catalogue number.

3.6 Manufacturer's Warranty

Provide manufacturer's warranty agreements where applicable.

Complete all necessary documentation on the Owner's behalf.

Include copies of warranties in the maintenance manuals.

3.7 Submissions

3.7.1 General

Submit drawings, samples, technical data or other details as applicable for systems as specified, prior to ordering or installation.

Allow sufficient time for checking and return of submissions.

Co-ordinate submissions of related items.

3.7.2 Drawings

Provide hard copies as follows:

- A1 size or larger – 3 sets
- A3 size – 1 set
- Electronic - PDF

Drawings shall be submitted in original contract size, AutoCAD or REVIT format as applicable.

3.7.3 Samples

Provide samples labelled to indicate type, part or catalogue number.

3.7.4 Technical Data

Provide manufacturers technical specifications, type test report, performance data, installation details etc. as required.

3.7.5 Co-Ordination

Co-ordinate shop drawing design with other building and services elements.

3.8 Certification

Certify that the works have been installed to the requirements of all applicable rules and regulations.

Provide a Form 16 at completion of the works.

3.9 Hazard Assessment and Risk Management

Comply with the requirements of the applicable State Occupational / Workplace Health and Safety Act and Regulations with respect to hazard identification and risk management.

3.10 Electromagnetic Compatibility

Ensure that the entire installation covered by this specification meets the Electromagnetic Compatibility requirements of the Australian Communications and Media Authority.

3.11 Packing Storage and Protection

Plant equipment, apparatus, materials and parts shall be delivered to the site in as new condition and properly packed and protected against damage due to handling or adverse weather or other circumstances.

Arrange suitable storage areas for materials, plant and equipment. Storage shall be on purpose made racks, mounts, etc., to ensure no damage or deformation of items.

Any items suffering damage in transit or whilst on site shall be rejected and replaced without cost to the Contract.

3.12 Setting Out

The positioning of equipment as shown on the Drawings is approximate and diagrammatic only, and these Drawings shall not be used for exact setting out. Locations shall be confirmed prior to installation.

Set out detector and sprinkler locations to comply with the relevant standards. Do not scale off drawings for any equipment locations.

3.13 Installation

Install equipment and services square and in line with the building elements, plumb and at consistent mounting heights.

Where possible fix directly to the building structure.

Provide noggings, brackets, fixings and supports to manufacturer's recommendations and as required.

Where timber is used for supports or noggings, provide Cyprus pine or similar termite resistant species.

Organise reticulated services neatly and with applicable segregation.

Provide separate cable support systems and support all cables over their entire route length.

Provide all access equipment and temporary lighting necessary for the installation of the fire protection services.

Allow for supports suitable for seismic (earthquake) restraint to AS 1170.4 where applicable. Refer to Clause 3.18.

3.14 Co-ordination

Liaise and co-ordinate with all equipment suppliers to ensure that all equipment is installed and connected in accordance with the installation and functional requirements of the manufacturer. Provide ancillary equipment as required for correct system operation.

3.15 Seismic Restraint

Provide equipment and equipment installation suitable for seismic restraint as required by NCC/BCA Sections BP1.1 and B1.2. The reference standard is AS 1170.4 Earthquake actions in Australia, and the most relevant section is Section 8 Design of Parts and Components.

For this specification, the seismic restraint pertains in particular to:

- Support brackets
- Fixings and fasteners, especially fixings into concrete and masonry
- Cable trays and ladders and the attached cables, especially long runs where differential building movement could cause cable damage
- Heavy equipment, especially where installed at high level relative to ground level, where installed overhead and where installed in ceiling spaces
- Pipe runs subject to differential movement of building elements

- Fixings for electrical switchboards
- Equipment associated with safety systems and smoke control systems
- Equipment and plant mounted on vibration isolating mounts

Provide Form 15 Design Certification and Form 12 Installation Certification by appropriately qualified and licensed structural engineer.

3.16 Environmental Conditions

Ensure that all equipment is suitable for the environmental conditions to be encountered under normal circumstances for the site.

3.17 Maintenance Access

Locate all relevant equipment to ensure adequate maintenance access in accordance with Workplace Health and Safety regulations.

3.18 Corrosion Protection

All metallic materials and components shall be suitably protected against corrosion.

Bolts, screws, rivets, etc for external use shall be non-ferrous or stainless steel, or where approved galvanised steel.

Brackets, plinths, rods, etc shall be hot dip galvanised after fabrication.

All steel sheets shall be corrosion protected by means of degreasing, priming and painting as a minimum.

Where the application is in a highly corrosive environment such as in close proximity to the sea, provide suitable grade stainless steel or non-ferrous metallic components.

3.19 Finishes

All equipment shall be painted unless the equipment material is chromium, anodised aluminium, plastic, stainless steel or other non-ferrous metals.

Paint and paint application shall comply with AS 2311 and AS 2312.

Carry out all powder coating to AS 3715.

3.20 Adhesives

Provide adhesives and sealants capable of transmitting imposed loads, sufficient to ensure the rigidity of the assembly or the integrity of the joint and which will not cause discolouration or lack of adhesion of finished surfaces.

3.21 Fasteners

Provide masonry anchors, bolts, nuts, washers, screws, nails and plugs as required for installation of the fire protection services.

Fastener materials, finishes and physical sizes shall be as appropriate for the loads, the types of materials in contact with the fastener and the environment.

Fasteners shall be selected to ensure full compliance with the requirements of Clause 3.18 Seismic Restraint.

3.22 Metals

Provide metal sections and sheet as required.

Finish invisible joints by welding, brazing, soldering, grinding and buffing prior to application of finishes such as galvanising or painting.

Repair galvanised or electroplated metals using two pack organic primer to AS 3750.9.

3.23 Labelling

Provide labels to equipment as required for identification, operation and warnings.

Labels shall be non-ferrous metal, multi-coloured laminated plastic or pre-printed flexible plastic.

Labels shall be of proven longevity without significant deterioration.

Externally mounted labels shall be weather and corrosion resistant.

Fix labels using nuts and bolts, rivets or self-adhesive material for flexible plastic labels.

Externally mounted fixings shall be corrosion resistant.

Warning labels shall be white lettering on red background with minimum 10 mm high lettering.

Minimum letter heights shall be 3mm.

3.24 Building Penetrations

Maintain integrity of building systems such as waterproofing and termite management at building penetrations, in particular:

3.24.1 Fire Integrity

Where systems pass through a fire rated building element, seal around the penetration in accordance with AS4072.1 to ensure that the required Fire Resistance Level of the building element is maintained.

Comply with the Building Code of Australia statutory requirements.

Label all penetrations through fire rated building elements and provide a penetration schedule and drawings in accordance with AS1851.

3.24.2 Smoke Barriers

Where systems penetrate smoke barriers, install and support this equipment to fully comply with all statutory requirements for smoke barrier building elements.

Label all smoke barrier penetrations and provide a penetration schedule and drawings in accordance with AS1851.

3.24.3 Acoustic Integrity

Seal all penetrations through acoustic rated building elements to ensure that the acoustic integrity is maintained.

Provide a detailed schedule of each acoustic rated penetration including the method of sealing and the location of each penetration. Provide copies to the Main Contractor for compilation into complete schedule of all acoustic rated penetrations.

3.25 Operation and Maintenance Manuals

Provide Operation and Maintenance Manuals for all systems and components covered by this specification. Include As-Installed documentation for the entire installation in these manuals.

Provide two copies of manuals and drawings neatly bound into A4 size hard cover ring binders which are permanently labelled with the Project Name, Contract Description and Contractor's Name. Provide full electronic copies of manuals in each of the hard copy manuals, either on CD or on USB memory stick. Electronic copies shall contain all manufacturer's information, as-installed drawings in REVIT, CAD and PFD format, forms, commissioning results, etc. as listed below.

Include the following in each set of manuals:

- Index
- Copies of all test results, certificates and approvals.
- Manufacturer's operating instructions and details of model numbers, serial numbers, operating instructions, and manufacturer's guarantee certificate for equipment and appliances.

- Logbooks to AS 1851 requirements
- Brief description of the installation
- A recommended maintenance programme for all equipment installed.
- A recommended list of spare parts.
- A copy of the Baseline Data documentation.
- Paper prints of all As-Installed Drawings as follows:
 - Layout drawings in the same size and form as the original contract drawings but modified to reflect all changes which occurred during construction.
 - System schematic diagrams and schedules.
 - The title block of each drawing amended to include the Contractor's name and contact details, Contractor's drawing number and date of issue.
- A portable electronic storage device such as USB stick, flash drive or CD Rom containing electronic copies of all As-Installed drawings in AutoCad or REVIT format as applicable, as well as in PDF format.

Provide a draft set of the Operation and Maintenance Manuals to the Superintendent not less than 2 weeks before the date of practical completion, for the purpose of review and comment.

Submit final sets of the Operation and Maintenance Manuals to the Superintendent not more than 2 weeks after the receipt of comments on the draft from the Superintendent.

Final contract payment will not be approved until satisfactory Operation and Maintenance Manuals have been received.

3.26 Clean Up

At completion clean all equipment to as-new condition.

Remove all waste and debris on a continuous basis.

3.27 Completion

All equipment and systems shall be fully operational and complete, all in accordance with manufacturers recommendations, prior to final testing and commissioning.

3.28 Testing and Commissioning

Fully test and commission the complete installation prior to practical completion.

Provide project specific Inspection, Testing and Commissioning Plans for all systems. Only suitably qualified personnel shall carry out testing and commissioning. Provide evidence of qualifications when requested.

Record all test results and submit for inspection prior to practical completion.

3.29 Construction Phase Procedures

3.29.1 General

Comply in all respects with procedures established by the Superintendent for the construction phase of the project.

3.29.2 Line of Communication

Ensure compliance with formal transmission of emails, RFIs and submissions through the agreed line of communication.

Advice received by other avenues will not be considered as a formal instruction.

3.29.3 Variations

Variation price submissions shall be accompanied by a detailed description of the work involved. Provide supporting documentation where necessary and as requested.

All costs shall be submitted on a spreadsheet showing itemised break-ups of all materials and labour.

3.30 Practical Completion

Practical Completion will be granted after the following is provided:

- Finalisation of all testing and commissioning
- Certification that the installation complies with the specified requirements and all applicable Statutes and Standards
- Authorities approval obtained as required
- Preliminary instruction on safe plant operation has been provided
- Critical defects have been rectified
- Operation and Maintenance Manuals and As-Installed drawings have been provided.

3.31 Test Instrument Calibration

All test instruments shall have current calibration certificates. Certificate copies shall be provided to the Superintendent prior to commencement of tests. Certificate currency shall be 12 months maximum.

3.32 Inspection at Practical Completion

Arrange for a Practical Completion Inspection by the Superintendent. A minimum of 5 working days' notice shall be given.

Ensure that the installation is fully operable and practically completed before requesting an inspection.

Should the inspection reveal that the installation is not fully operable or practically completed and require further inspections, these inspections will be charged to the Contractor.

3.33 Instructions and Training

Provide skilled operators to instruct such persons as may be nominated by the Superintendent in the effective operation of all of the fire protection systems in the installation. Instructions shall not be given until completion of all tests and adjustments.

Conduct training at agreed times and locations and for durations commensurate with the complexity of the systems.

3.34 Maintenance and Warranty During Defects Liability Period

Carry out preventative maintenance on a regular schedule for systems as detailed in other sections of this specification.

Promptly rectify all defects during the defect's liability period. Attend site within 1 working day of notification of defect. Critical system defects shall be rectified in the time schedule as detailed in other sections of this specification.

4. FIRE DETECTION SYSTEM

4.1 General

4.1.1 System Design and Operation

Perform the detailed design and documentation of the project. Carry out a design check and verification to ensure that the completed installation complies in all respects all relevant regulations, codes and standards.

Determine logical addressable loop layout and fire alarm zones based on building layout and configuration.

Design the building occupant warning system to comply with the requirements of NCC Volume One Section E which references AS 1670.1 Clause 3.22. The occupant warning system shall comply with AS 1670.1 Clause 3.22 (a) or (b) as applicable and as required by NCC Clause E4.9.

Determine logical emergency evacuation zones generally based on building configuration, the fire emergency plan and the fire/smoke compartments.

Design evacuation zone signal stages (cascading sequence).

Obtain agreement with the Fire Authority for the unwanted alarm strategy and provide systems and equipment as required to comply with this strategy.

Submit design data on shop drawings.

4.2 Quality

4.2.1 Standards

Conform to all applicable standards, especially as detailed in Clause 3.2 as well as all standards referenced in AS 1670.1.

4.3 Products

4.3.1 Authorised Products

All provided equipment shall be listed in the ActivFire Register of Fire Protection Equipment.

4.3.2 Control and Indicating Equipment (CIE's)

Provide control and indicating equipment to AS ISO 7240.2 and AS 4428 Series only where applicable.

Provide FDCIE, DPCIE, EWCIE and EICIE as required for the fire alarm system.

All control and indicating equipment shall be open protocol, capable of being programmed, serviced and amended by independent fire services contractors. Closed protocol systems able to be serviced or altered by a single company only will not be acceptable.

Ensure adequate access for control and maintenance, as detailed in AS 1670.1.

Locate all CIE's in metal cubicles constructed to manufacturers standard construction.

Batteries shall be housed in integral but separate compartments, the interior of which shall be corrosion protected.

Install Fire Authority equipment as required for system operation.

All CIE shall leave spare capacity for future additions as follows:

- 50% spare zones in FDCIE
- 30% spare capacity on each addressable loop
- 30% spare capacity in sound system and intercom system panels and equipment
- 50% spare space in battery compartments.

Provide unwanted alarm mitigation equipment as required by the Fire Authority, alarm signalling equipment to AS 4428.6 and fire fighters interface to AS 4428.3, and all to Fire Authority requirements.

Provide power supply equipment conforming to AS ISO 7240.4.

4.3.3 Zone Block Plan (Locality Diagram)

Provide zone block plan of minimum A3 size and as necessary to clearly indicate items as detailed in Clause 3.10 of AS 1670.1.

The panel shall be rear engraved coloured plastic in a suitable mounting frame.

4.3.4 Detectors

Provide universal base analogue addressable detectors.

Detectors shall incorporate LED indication of operation status and alarm condition.

Provide point type smoke detectors generally photo-optical type to AS ISO 7240.7. Ionisation type smoke detectors to AS ISO 7240.7 shall also be provided as required by regulation or to suit the application.

Heat detectors shall have a temperature duty range to suit the application. Provide heat detectors shall complying with AS ISO 7240.5.

Provide multi sensor detectors complying with AS ISO 7240.15.

Provide linear cable type heat detectors complete with mounting brackets, catenary wires, stainless steel cable ties, test points, junction boxes and insulating pads as required.

Provide air duct sampling detectors purpose built for the application and to AS 1603.13.

4.3.5 Mountings for Concealed Space Detectors

Provide suitable mounting facilities to enable maintenance access for all concealed space detectors.

Mounting facilities shall incorporate suitable lock-in arrangement to ensure correct detector location at all times.

4.3.6 Remote Indicators

Provide remote indicators to AS 1603.15.

4.3.7 Visual Alarm Devices

Provide VAD to AS ISO 7240.23

4.3.8 Building Occupant Warning System

Provide an occupant warning system complying with AS 1670.1, Clause 3.22 as follows:

- An occupant warning system with evacuate signal only and complying with AS 1670.1, Clause 3.22.1 (b), either by EWCIE and speakers or audible alarm devices conforming to AS ISO 7240.3 as applicable.

4.3.9 Loudspeakers

Provide sound system loudspeakers to distribute audible signals in accordance with AS 1670.1 and AS 1670.4 as applicable.

Speakers shall be 100mm diameter, recess mounted, dual cone, minimum 3 watt rated, adjustable tap settings with clip on white grille, complying with AS ISO 7240.24.

4.3.10 Horn Speakers

Provide sound system horn speakers complying with AS ISO 7240.24 for exterior, exposed and plant locations.

Horn speakers shall be weatherproof, moulded high impact plastic case, 10 watt rated and with adjustable tap settings.

4.3.11 Visual Warning Devices

Provide visual warning devices complying with AS 1603.11 and AS 1603.17 where required due to anticipated ambient noise conditions and hearing impairment.

4.3.12 Interfaces

Provide hardware, software, cables, terminations and peripheral equipment as required for high and low level interfaces as detailed.

4.3.13 Cables and Cable Supports

Provide all required cables and cable support systems including trenching and backfilling for underground conduits.

Provide cable types as recommended by the fire detection system and emergency warning / intercom system equipment providers to ensure full functionality of the system.

Provide fire rated cables as required to comply with the relevant standards.

Provide fibre optic external grade cable for inter-building connections.

Provide fibre optic media converters, fibre terminations and all associated racks and cabinets as required.

4.3.14 Surge Protection

Provide lightning and surge protection filters on all CIE and inter-building copper cables at both ends.

CIE surge protection shall comprise separate proprietary surge filters connected to the CIE 240 volt power supply. Surge filters shall be Erico Critec Transient Surge Filter, 6A or 20A rated as appropriate.

4.4 Installation

4.4.1 Fire Alarm Monitoring

Connect system to a monitoring service provider (or fire brigade as appropriate).

Separate signals shall be transmitted for the fire detection system and the sprinkler system.

Connect system to the monitoring service via wireless alarm signalling equipment to fire authority requirements.

Install fire rated cable from FDCIE to the Campus Distributor and terminate as a backup hard wired system.

Arrange via the Proprietor and Superintendent for application and payment to the Communications Carrier for the monitoring connection.

4.4.2 Control and Indicating Equipment

Install all CIE's in locations shown.

Ensure provision of required access and maintenance space to AS 1670 Series.

Provide high level interfaces between separate CIE panels.

4.4.3 Zone Block Plan

Install on wall adjacent to FDCIE at the Designated Building Entry Point (DBEP or DSEP as appropriate).

4.4.4 Fire Fan Controls

Run all cables and install controls as required from the FDCIE/fire fan control panel to the mechanical plant as required to comply with AS 1668.1.

Co-ordinate requirements and interfaces with mechanical services.

4.4.5 Detectors

Install detectors clear of other services and equipment and at spacings to comply with AS 1670.1.

Label all detectors to indicate address and loop numbers.

Install remote indicators and provide suitable maintenance access provisions for concealed space detectors.

Install sampling detectors within air ducts in locations co-ordinated with mechanical services.

Install line type heat detectors on brackets and catenary wires in accordance with manufacturers recommendations.

Install smoke alarms to comply with AS 1670.6 and NCC Volume One.

4.4.6 Accessibility for Maintenance and Testing

For all equipment requiring maintenance and/or testing install so that it is readily accessible based on the applicable work place health and safety regulations.

4.4.7 Building Occupant Warning System

Install zoned building occupant warning system speakers and visual warning devices in suitable locations.

Determine zone labelling and descriptions in conjunction with the Superintendent and Owner.

Co-ordinate programming of high level interface between CIE's.

Adjust output levels of all speakers and amplifiers to provide optimum alarm signal pressure levels and to ensure distortion free voice messages.

4.4.8 Cables and Cable Support Systems

Install all required cables and cable supports in accordance with manufacturer's standard requirements and the following clauses.

Select transmission paths for cable routes to provide the maximum mechanical protection and physical security for the cabling systems.

Provide transmission path protection in accordance with Clause 3.26 in AS 1670.1.

All wiring systems shall be fully supported over their entire route length by means of cable trays, ducts, conduits, catenary wires etc.

Wiring shall be completely segregated from 240V ac power and other wiring systems and run on/in separate cable support systems.

Fire rated cables shall be supported to comply with the requirements of the manufacturer to ensure cable integrity to match the certified fire resistance.

All wiring shall be installed on the loop-in system with connections made at outlets. The use of junction boxes shall not be permitted.

All wiring shall be installed concealed throughout except in plant rooms, stores and the like where cables shall be run concealed in conduit.

Cables shall be supported clear of the ceiling and shall not be laid direct on the ceiling.

All conduits shall:

- be 20mm minimum diameter;
- be set at all changes of direction where concealed within the structure;
- be fitted with a sealed cap to prevent the ingress of moisture or obstructions entering the conduit;
- be fitted with well-designed expansion joints when conduit runs between slabs of concrete deliberately separated for thermal movement;
- be positioned with a minimum of 25mm of cover within concrete structures.

Surface mounted conduits shall be rigid non-metallic conduits complying with AS 2053 and shall be fastened at intervals not exceeding 900mm using single or double sided saddles as directed and fixed to approval. Corrugated conduit shall not be used.

Conduits shall be laid in straight and symmetrical lines with easy sets or bends. Bends or sets must be made without altering the section of the conduit.

Where it is not practicable to set the conduit, large sweep bends may be used.

Join PVC conduits in accordance with the manufacturer's instructions using the recommended jointing cement.

Conduits concealed in floor slabs and walls shall be rigid non-metallic or corrugated non-metallic conduit complying with AS 2053 and, where installed in screeds and chases, shall be secured at intervals not exceeding 1800mm with approved fixings. Conduit stub-ups through concrete surfaces and foundations shall be adequately braced to prevent shifting during pouring and shall be adequately plugged to prevent ingress of dirt.

4.4.9 Underground Systems

4.4.9.1 General

Provide excavation, removal from site, rock excavation, conduits, pits, backfilling and reinstatement for all underground cabling.

Cables shall not be installed direct buried.

Cables run in underground conduit systems shall be exterior grade.

Record the route of underground systems prior to backfilling.

Set out conduit routes and obtain approval for these routes prior to excavation.

Provide polypropylene draw cords or insulated copper draw wires in all spare conduits.

Allow for final landscaped and finished levels when installing conduits and pits.

Maintain minimum required cover. Pit lids shall be installed flush with the surrounding finished levels.

4.4.9.2 Inspection and Hold Points

Each trench section must be inspected for approval by the Superintendent prior to inspection.

A photographic record may be acceptable for this purpose but it will be necessary to obtain Superintendents approval for this prior to proceeding past the hold points.

Hold points shall be:

- When the trench excavation is completed
- When the services have been laid in the trench

Arrange for inspection by the Superintendent at each hold point.

4.4.9.3 Excavation

Ascertain the locations of all other in-ground services prior to excavation.

Saw cut or remove existing surface as appropriate.

Excavate trenches to the required depths and remove excess material from site.

Keep trenches free of water. Place bedding material, services and backfilling on firm ground.

Remove all sharp objects, rocks and organic material.

Give notice if rock is encountered to enable agreement on quantity.

Provide safety barriers to open trenches.

Excavate and fill trenches in short sections to minimise safety hazards.

4.4.9.4 Boring

Engage a suitably qualified specialist contractor for all boring works. Fill all voids by pressure grouting.

4.4.9.5 Conduits

Provide bedding sand below, around and above conduits with minimum 50mm cover. Bedding sand under roads shall be stabilised with 3% cement mix.

Maintain separation between fire, electrical, communications, Telstra, gas, water and other underground services to statutory requirements.

Use large radius bends for turned up conduits.

Use large radius bends and/or pits for changes of direction.

Neatly cut conduits and remove all burrs and conduit filings so that cable stripping is avoided.

Clean installed conduits free of debris and water.

Seal all conduits after cable installation using a product specifically manufactured for this application.

4.4.9.6 Pits

Provide sufficient pits to enable easy installation of cables.

The top of pits shall be finished flush with the surrounding surfaces.

Prior to installation, cut pit walls for conduit entries using separate circular holes for each conduit. Use suitable hole saws for this work. Pits cut in large sections for more than one conduit entry shall be rejected and pits shall be replaced.

Seal around conduit entries to after pit installation to prevent ingress of water and silt.

Drain pits to stormwater drainage system. Alternatively, obtain approval for rubble absorption trenches if draining to stormwater is not possible. Minimum rubble trench size shall be 300 x 300 x 2000 long using coarse gravel.

Install pits into a levelled and compacted excavation.

Found pits onto a wet concrete base at least 50mm thick or onto a compacted rubble base where the pit is drained to rubble.

The top of the pits shall finish flush with surrounding surfaces.

Cut conduit entries not more than 50mm past the inside faces of the pit walls.

Provide rigid conduit bracing bars inside the pits to prevent wall collapse.

Provide poured concrete haunches (minimum 150mm thick) around multiple conduit entries.

For pits in trafficable areas (Class C and above) backfill around the pits to the finished surface with concrete (minimum thickness 100mm).

For all other pits backfill around pit using compacted bedding sand and finish at the surface perimeter of the pits with a 200 wide and 200 thick trowelled concrete collar, reinforced as necessary.

- 4.4.9.7 Backfilling
Backfill trenches and compact using a ramming machine in maximum 200mm layers. Complete the backfill with a 100mm minimum layer of topsoil.

Backfill material shall not contain any rocks, organic material, or sharp objects.

Backfill under roads and paved areas shall be fine crushed rock compacted in 150mm layers to 100% STD compaction.

- 4.4.9.8 Reinstatement
Existing surfaces shall be reinstated to their original condition.

Lawn areas – Reinstatement removed lawn segments or provide new turf to match existing.

Paving – Lay pavers on sub-base to match existing. Provide additional pavers as required.

Concrete – Key new poured concrete into adjacent existing concrete using reinforcing steel grouted into drilled holes.

Roads – Provide rock sub-base, base and bitumen to match existing.

4.4.9.9 Route Markers

Provide route markers to accurately indicate the location of underground conduit routes.

Install markers on pits, at 50 m intervals on straight runs, at changes of direction and at ends of run.

Markers shall consist of 16mm thick engraved stainless steel arrows, discs or rectangles as appropriate showing direction and type of service. Fix markers to concrete base using stainless steel friction pins with mushroom head.

4.4.9.10 Cables

Install cables within conduit systems in accordance with manufacturer's instructions.

Cables with sheathing damaged during installation shall be removed and replaced.

4.4.10 Interfaces

4.4.10.1 Sprinkler System

Provide interfaces for sprinkler system alarm (flow switch) and valve monitoring. Run cables to the sprinkler control valves and connect.

4.4.10.2 Mechanical Services

Provide shutdown alarm signal to mechanical services switchboard fire relay and terminate. Co-ordinate details with mechanical services or fail-safe operation.

Run cables as required and terminate at fire alarm relay.

4.4.10.3 Electronic Security System

Provide fire alarm signals on a zone by zone basis for fire release of secured doors. Co-ordinate requirements with security services and run cables as required to security panels.

4.4.11 Programming

Carry out all required data entry and system programming.

Obtain room names and codes from the client representative prior to programming to maintain consistency of room names and codes throughout.

Confirm cascading sequence details prior to programming the system.

Install all equipment as required for a completed, fully operational system.

4.5 Testing

4.5.1 On Completion

Carry out tests, including out-of-hours tests, to demonstrate the automatic fire detection and alarm system's performance to AS 1670.1 and the compliance Sections of the relevant parts of the AS 1603 series. Include the following:

- Test components for correct function and operation.
- Demonstrate detection and alarm performance on site, to at least the level stated in the manufacturer's performance specification for that device.
- Test alarm zone identification.
- Demonstrate addressable device operation for 14 days with data logger to verify stability of detectors and devices.
- Test interface to interconnected systems.
- Demonstrate correct shutdown sequences during fire mode.
- In situ testers shall comply with AS 1603.16.

4.5.2 Weekly

After the installation has been successfully acceptance tested, instruct the Superintendent on how to carry out weekly tests of the fire protection installation, in accordance with the testing requirements of AS 1670 and AS 1851.

4.5.3 Monthly

After the insulation has complied with the acceptance test requirements, monthly tests, in accordance with the maintenance and testing requirements of AS 1670 and AS 1851, shall be carried out during the specified maintenance period after the date of practical completion.

4.5.4 Annual

Complete an annual test of the complete system one month before the end of the maintenance period.

4.6 Completion

The whole of the installation shall be left complete in every way. After necessary tests required by the Fire Brigade have been carried out, the written approval of such Authority shall be handed to the Superintendent.

4.6.1 Training

Provide on-site training for staff in the operation and testing of the alarm and warning systems. Allow 2 sessions each minimum of 1 hour duration.

4.6.2 Maintenance

Carry out maintenance and complete records to AS 1851 for the duration of the defects liability period.

4.6.3 Clean Up

Leave all equipment in “as-new” condition.

Remove all waste from site.

4.6.4 Baseline Data

Provide baseline data to the proprietor in accordance with the requirements of AS 1670.1 and AS 1851.

Leave a full set of A3 1:200 or A1 1:100 printed as-built drawings for all services that have interfaces to the fire alarm system. Leave drawings inside the fire panel on completion. All devices shall be numbered on the as-built drawings to match the labels on the devices in the field.

Obtain a copy of the most current Fire Engineering Report (FER) and provide a bound A4 copy in the fire panel

Include bound, printed copies of all fire systems design and installation certificates and reports, fire brigade approvals, building certification.

Certificates, reports and as-built drawings shall clearly indicate all required standards that the system complied with at the time of completion, including the revision/version of each standard, to ensure future fire system maintenance service providers are clear on what standards the system is required to maintain compliance with.

In addition to the printed documents, provide a USB drive with PDF copies of all documents listed herein. Leave this USB in the fire panel enclosure.

Where the documents listed cannot easily fit within the fire panel enclosure, provide an additional wall mounted enclosure below, above, or adjacent to the fire panel enclosure to house the documents in a neat, logical manner. The box is to be powder-coated steel, red in colour, keyed with a 003 key, and provided with suitable sleeves to house and protect the documents. The box shall be labelled clearly.

Provide copies of and references to all documents listed herein, in the Fire Alarms section of the project operations and maintenance manual in accordance with AS 1670.1 and AS 1851.

5. FIRE SPRINKLER SYSTEM

5.1 Design Criteria - Fire Sprinklers

The fire sprinkler system shall be designed and calculated in the following manner:

Design the automatic fire sprinkler protection to the premises to comply with the requirements of AS2118 Part 1:

- Ordinary Hazard 3 for the building areas as well as under the awnings external to the buildings
- Light Hazard for the office and amenities areas in the buildings

Coordinate the proposed design layout with all other services layouts and the architectural/structural drawings.

Calculate the final pipe sizes. Pipework sizing of the main connection shall allow for future extensions.

Carry out the hydraulic calculations using the “Hyena” computer analysis programme.

Calculate the final fire pump duty requirements prior to ordering the pumps. NOTE that pumps are combined for hydrant and sprinkler duty, liaise with hydrant services contractor.

Calculate the final water storage requirements prior to ordering the tank. NOTE that pumps are combined for hydrant and sprinkler duty, liaise with hydrant services contractor.

Submit designs, calculations and shop drawings for approval prior to fabrication of piping.

5.2 Sprinkler Heads

All sprinkler heads shall be on an approved type, with orifice sizes and deflector patterns to suit the specified risk areas.

Sprinkler head type shall be as follows:

- Semi recessed pendant, 68 degrees Celsius, fast response, white with white escutcheon to suit ceiling finish.
- 10 mm brass, 79 or 93 degrees Celsius, fast response glass bulb type pendant/upright in ceiling concealed spaces where indicated on the drawings.
- 15 mm brass, 68 degrees Celsius, fast response glass bulb type in exposed areas. 93 degree C under awnings.
- Specific dry heads suitable for use in freezer areas

5.3 Valves and Fittings

5.3.1 Alarm Valves

Provide full way swing type. The alarm valve set complete with 50mm flow test drain shall be installed where indicated on the drawings.

5.3.2 Pressure Switches

Provide totally enclosed, waterproof type, rated to at least 1 A to operate on a drop in installation pressure to initiate the alarm condition indication to the FIP.

5.3.3 Fire Sprinklers Alarm Gong

The fire sprinkler alarm gong shall be located where indicated on the drawings and comply with AS4118 Part 1.3. The gong shall be activated upon the operation of a single sprinkler head. Gong wastes shall be run to discharge in the floor waste within the control valve cupboard.

5.3.4 Remote Test Valves

Provide 25 mm globe or ball remote test valves, piped to discharge to ground level, at the extreme points of the sprinkler system as indicated on the drawings to meet the requirements of clause 8.11 of AS2118 Part 1.

5.3.5 Fittings

Use fittings with wall thickness to match that of the pipe.

Screwed fittings shall be heavy pattern, low carbon steel suitable for making screwed joints to AS1722.1.

Formed (pulled) bends shall be to AS4041.

Elbows less than DN 50 shall be screwed type or rolled groove couplings.

Mechanical roll-grooved fittings shall be used only in accessible locations, in sprinkler protected areas.

Bolts for flanges shall be made out of carbon steel to AS2129 Appendix C.

Fittings shall be zinc plated in noncorrosive environment, otherwise a protective finish with equivalent corrosion resistance to, and compatible with, the flange.

5.3.6 Drain Valves

Connect the sprinkler system test and auxiliary drain valves to the nearest convenient points of the drainage system as indicated on the drawings.

5.3.7 Locking

Padlock drain and test valves in the closed position.

5.3.8 Flow Test Facility

Provide at the pump set, an appropriately sized flow test device which shall be complete with regulating valve, stop valve and flow test element. The drain shall be run to terminate into the high level water return connection on the sprinkler water storage tank adjacent to the pump enclosure.

5.4 Pipework

5.4.1 General

All pipework shall be medium grade black steel tube complying with AS2118 Part 1 and Part 9, except where external to the building envelope where it shall be hot dip galvanised and painted in accordance with general specification painting clauses.

The drawings indicate the manner in which the piping is to be run and the nominal location of risers.

Size all pipes in accordance with AS2118. No extra payment shall be allowed for incorrect sizing.

All work, whether shown on the drawings or specified, shall be installed in an approved manner to meet structural, architectural and other site/or service conditions.

Construct and install pipework so that it is free to contract or expand without damage to itself or any other part of the building.

Special care shall be taken in the arrangement of piping to ensure neat and workmanlike appearance and true alignment and grade. All pipes and fittings shall be thoroughly cleaned before erection, removing all scale, burrs, fins and obstruction and each section shall be blown out to remove all scale, slag, pipe cutting residue, etc, prior to the final installation.

All levels and heights of pipework shall be detailed on the shop drawings submitted for approval by the Contractor.

Pipes shall be joined by threaded end, screwed sockets, mechanical couplings and screwed fittings.

Threaded pipe joints shall be sealed as follows:

- Hemp in conjunction with an approved sealing compound;
- Teflon tape, or
- Liquid Sealant "Avdel Permabond A 129" or approved equivalent.

The assembled coupling shall fully enclose the gasket within the housing of the coupling.

All galvanised malleable cast iron pipe fittings shall comply with BS1256 and shall be suitable for threading as specified as AS1074.

On small diameter pipework, plain end locking type fittings as manufactured by Victualic Company or similar approved may be acceptable. However, prior approval must be obtained for this contract from the Authority having jurisdiction by the Sub Contractor. The Superintendent shall then advise if this type of fitting is permitted.

5.4.2 Pipe Supports and Fittings

All pipe hangers, supports and fixings shall conform to the requirements of AS2118 Part 9 "Piping Support and Installation".

Where mechanical fittings are employed, provide support within 600mm of both sides of the coupling.

5.4.3 Roll Groove Couplings

When grooved jointing coupling systems are installed, they shall be used only on pipe sizes up to 300DN. Submit complete technical details for approval prior to proceeding. Gaskets shall be of resilient elastomeric material conforming to the internal cavity of the housing and sealing against the pipe as the housing of segments are tightened. Housing shall be of ductile or malleable iron fully enclosing the gasket. Housing keys shall engage into pipe grooves around full circumference.

Contractors shall comply with the requirements of relevant Australian Standards.

The following installation conditions shall apply:

- the installing Contractor shall have significant experience with the product and be able to provide at least three references
- Contractors' staff shall have completed a one day training session for grooving of pipe and control of Quality records, conducted by the pipe manufacturer and grooved jointing manufacturer
- Contractors' staff shall have completed a one day training session for installation procedures for the grooved jointing coupling products, conducted by the pipe manufacturer and grooved jointing manufacturer
- the piping system design for hangers, anchors and expansions shall be reviewed by the grooved jointing manufacturer prior to commencement of any installation
- roll grooving shall be done with purpose made machines and roll sets
- ensure that the Manufacturer's Product Warranty cover is valid, only product manufactured by the single approved grooved jointing manufacturer shall be used
- employ the staff of the pipe manufacturer and grooved jointing manufacturer for onsite training and to periodically monitor the installation to ensure it follows the grooved jointing manufacturer's installation standards and procedures

- Contractor shall verify that all grooved jointing coupling joints are bolted up metal to metal at the pads.
- Galvanised fittings and couplings are to be inspected for uniformity of the zinc coating. Excess zinc that reduces the effective depth of the joint face zones (key zone) are to be rejected. Galvanising is to conform with AS/NZS 4680
- Contractor shall perform inspections of ALL roll grooves, both on-site rolled and factory deliveries, and maintain records for future verification.
- Couplings shall have manufacturers name as part of casting
- Couplings shall be batch stamped on inside of both halves for future traceability. Couplings that are not batch stamped shall be rejected. Style or model number only is insufficient information as is to be rejected.
- Rubber gasket shall form part of the coupling delivery, they shall not be a separate stock item, and they shall be pre-lubricated by manufacturer. Supplementary lubrication may be required however excessive lubrication may result in joint failure.

5.4.4 Pipe Fixings

Explosive fasteners shall NOT be permitted.

Fixing to timber shall be in accordance with AS2118 Part 9 Clause 2.5.3.

The manufacturer of the fasteners shall be consulted to determine the correct shank diameter and the correct fasteners to be used.

5.4.5 Pipe Hangers

Pipe hangers to brackets shall be properly isolated where necessary to prevent noise and vibration transmission.

Generally, pipe runs shall be grouped and spaced evenly and so arranged that a minimum number of crossovers are necessary.

Clamping shall evenly distribute bearing loads to the pipes and shall not impose bending moments.

All hangers, clips, clamps, brackets, fasteners etc., shall be of strong durable and fire resistant materials, and shall be installed in such a manner as to prevent excessive vertical and horizontal movement, sagging, vibration and strain on pipework joints and connections.

5.4.6 Materials

Use steel pipe in accordance with AS1074 medium wall thickness. Pipework external to the building envelope shall be hot dip galvanised.

5.4.7 Painting

All external pipework shall be hot dipped galvanised and left unpainted.

All internal exposed pipework shall be painted in accordance with following. Painting shall be carried out by this subcontract.

All paints shall be of the finest quality enamel paints. Paint shall be selected to withstand the temperature existing on the surface of the equipment to which it is applied.

The pipework installation shall be painted as follows:

- All exposed pipework – Surface preparation – Hand wire brushed with digressing etching agent, followed by:
- One coat of zinc chromate primer, and
- Two coats of Fire Brigade Red Oil and Petrol Resistant Enamel.

The colour of all exposed piping is to suit colour scheme as specified by the Superintendent.

5.4.8 Flexible Sprinkler Hose Assembly

As an alternative to the use of black steel pipe and galvanised malleable screwed fittings for sprinkler dropper fit off, a Stainless Steel, Braided type, Flexible Sprinkler Hose Assembly may be used.

The Flexible Sprinkler Hose Assembly shall be equal or equivalent to “Flexi-Dropper” as supplied by AFT Australia Pty Ltd and must be SSL/UL approved and listed, as well as listed on the ActivFire Register of Fire Protection Equipment. It must consist of a braided stainless-steel covered flexible metallic hose with a special inlet nipple and a special outlet reducer nipple, plus a ceiling grid support assembly suitable for securing to all types of ceiling grid.

The Flexible Sprinkler Hose Assembly shall have an identical female “slip-nut” at each end to facilitate connection to the supplied special nipples. Each “slip-nut” joint shall be fitted with an NBR ‘O’ ring to provide leak-free sealing. The “slip-nut” connection must be fitted to the Flexible Sprinkler Hose in a way to ensure the ‘O’ ring is secured in position and not exposed to avoid risk of ‘O’ ring falling off.

The inside diameter (ID) of the Stainless Steel Hose tube shall be a minimum 25mm. The hose tube shall be made from Grade 304 Stainless Steel Bright Drawn Finish. The outside braided sleeve shall be 0.3mm thick Grade 304 Stainless Steel Wire, Bright Drawn Finish.

The Ceiling Grid Support Assembly shall consist of a 1300mm length of mild steel bright drawn finish, square hollow section (SHS), two identical aluminium clamps for securing the SHS to the ceiling supporting channels, and a mild steel support bracket for rigidly clamping the outlet reducer nipple to the SHS. The bracket and clamps shall provide a

convenient and quick means of accurately setting-up the location of the sprinkler head relative to the ceiling grid. The aluminium clamps shall be a universal type capable of securing the SHS to all types of ceiling grid and suitable for both normal and fully concealed type sprinkler heads.

Flexible Sprinkler Hose Assembly must be installed in accordance with the manufacture's installation instructions provided. These instructions shall also include equivalent length details, determined in accordance with the requirements of UL1474.

Where Flex Droppers are proposed to be used, full hydraulic calculations shall be submitted showing the above equivalent length losses have been taken into account. Published equivalent length losses produced by the manufacturer shall be submitted with the calculations.

5.5 END-SUCTION CENTRIFUGAL PUMPS

5.5.1 Standard

General: To ISO2858 or DIN EN 733, and to the requirement of AS2914 and AS2118

5.5.2 Type

General: Provide single stage, radially split, end-suction, back pull out pumps, complete with pump controls in compliance with AS2914.

5.5.3 Performance

Minimum performance shall be as scheduled, plus the safety and additional requirement of AS2118 and AS2914.

Provide at two pumps with compression ignition engine.

5.5.4 Bases

General: Mount pumps and motors on bases sufficiently rigid to prevent distortion under normal operating conditions. Select from the following:

- Cold-formed bases: Mild steel plate minimum 6mm thick.
- Fabricated bases: Fully welded fabricated from hot rolled steel channel sections.

Inertia bases:

- Construction: Cast concrete with fabricated base with reinforcing bars welded between base sections, and with a combined weight at least that of the pump and motor.

Mounting brackets: Provide at least 4 height saving mounting brackets.

5.5.5 Mountings

General: Provide spring mountings at each bracket.

5.5.6 Casing

Pressure rating:

- For systems with a design pressure $\leq 500\text{kPa}$: Minimum static test pressure of 1.6MPa.
- For systems with a design pressure $> 500\text{kPa}$: Minimum static test pressure of 1.6 MPa or 1.5 x the total of the pump shut-off head plus the static and system pressures, whichever is the greater.

Material:

- Bronze, grade 836B.

Drip well drainage: Provide embossed, drilled and tapped drainage holes in seal drip wells. Drain to AS/NZS3666.1.

Jointing: Provide gaskets or O-rings, or both, compatible with the liquid to be pumped and shaped to prevent contact between the liquid and the casing fixings.

5.5.7 Connections

Piping connections:

- Screwed connections: Screwed female fittings to AS ISO7.1. Provide union on each connection. Do not provide screwed connections for working pressures $> 250\text{kPa}$, temperatures $> 100^\circ\text{C}$ or connections $\geq \text{DN}50$.
- Flanged connections: To AS 2129, minimum Table E or equivalent.

Fluid release: Provide embossed, drilled and plugged openings at the lowest casing points for drain cocks.

Pressure gauge tapings: Provide accessible, embossed, drilled and plugged openings integral with casings.

5.5.8 Rotating assemblies

Balancing: Statically and dynamically balance completed rotating assemblies.

Shaft: Solid grade 416 stainless steel, machined all over, free of stress concentrations.

- Maximum surface roughness: $25\text{ }\mu\text{m}$, under sleeves, seals and bearings.
- Maximum deflection:
- Torsional: 0.002 rad .
- Lateral: $1\text{ }\mu\text{m/mm}$ of shaft length.

Impellers:

- Type: One piece shrouded type with machined sealing collars.
- Material: Grade 836B bronze.
- Securing to shaft: Use a key and locking nut.
- Locking nut: Grade 316 stainless steel, or bronze.
- Parallel operation pumps: Matched impellers.
- Maximum diameter: 90% of maximum impeller size for the casing.

Shaft sleeves: Provide a sleeve extending the full length of the seal housing.

Thrower rings: Fit water thrower rings to shafts.

5.5.9 Couplings for direct coupled pumps

Couplings: Direct couple pumps and motors with flexible spacer couplings.

Spacers: For rear access pumps provide spacer couplings long enough to permit removal of rotating elements without disturbing piping connections or motor alignment. Provide a spigot for the flanged joint between the coupling and the spacer.
Coupling guards: Enclose couplings in coupling guards.

5.5.10 Bearings

General: Provide at least 2 deep groove ball bearings, widely spaced and selected for a minimum rating fatigue life of 17 500 hours.

Lubrication: Provide either grease or oil lubrication. Seal bearings against ingress of dust and moisture, with lip seals. Allow for release of excess lubricant. Provide grease nipples for grease lubrication.

5.5.11 Shaft seals

General: Provide seals compatible with the working conditions, including temperature and pressure of the fluid being pumped.

Type:

- Mechanical seals: Provide carbon elements rotating against a ceramic stationary face.
- Packed gland seals: Provide glands long enough to seal the fluid being pumped. Provide lantern rings if recommended for the particular duty.

Drains: Provide a drain from shaft seal well on pumps fitted with packed gland seals and all pumps on open water systems. Run drain to fall continuously to building waste.

5.5.12 Motors – Compression Ignition Engine

General: Provide compression ignition (diesel) engine compatible with pump requirements, giving efficient, non overloading pumping units.

Minimum power rating: At least the maximum power required by the pump when projecting the system resistance curve to the maximum impeller size for the pump casing size.

Fuel: Provide a full tank of diesel fuel upon hand-over of the system.

Microbial control: Provide initial treatment to the fuel prior to hand-over, provide documentation to proof that this has occurred and detail what has been used and needs to be used for continued treatment.

5.5.13 Installation

General: Adjust mountings so that units are level. Following connection and filling of piping, and before operation, align pumps and motors.

Maximum misalignment: < 50% of the manufacturer's recommended maximum.

Packing: Minimise the number of packers and shims.

Dowelling: Dowel pumps and motors after a running in period. Fix the position of the pump with dowels at diagonally opposite feet and, following final alignment, dowel the motor at diagonally opposite feet. Provide parallel type dowels.

5.5.14 Piping

General: Support pipes independently of pumps.

Disconnectable connections: Provide flanges or screwed connections with unions, for removal of pump casings without disturbing piping.

Suction connections: Provide demountable pipe sections between pumps and system isolating valves, for removal of impellers.

Pumps with packed gland seals: Drain drip well to the nearest waste with DN 25 copper pipe.

5.5.15 Controller

General: Provide controls in cabinet on pump frame in one assembly. Control functions, alarms, indicator lights, light test facility, etc. in compliance with AS2941 requirements. Pressure switches shall be located on the pressurised piping running to the buildings, downstream from the non-return valves above the pumps.

Provide outputs for connection to the BMS System:

- "Fault" alarm for each pump
- "Run" indication for each pump

5.6 Sprinkler Water Storage Tank

Allow all costs to supply and install as shown on the drawings the sprinkler water storage tank, complete and in accordance with AS 2304, AS 2419, AS 3500, local fire brigade and local authority requirements.

The tank shall be installed and consist of the following minimum requirements:

- Allow for external colourbond tanks. Exact colour to be determined by client prior to ordering of tank.
- Quick-fill valving & pipework for connection by Hydraulic Services
- Pilot-operated float valving equal to "Bermad"
- Water storage requirement: Sprinkler water full capacity (60 minutes)
- Suitability designed and installed structural base & bolted panel tank walls that shall be supported in such a manner that no load is transmitted to any of the attached pipework. Complete with minimum 25-year durability guarantee.
- Installed with a suitable tank liner that suits the application & applicable standards. Complete with minimum 10-year durability guarantee.
- Tanks shall be accessible for inspection, repairs & maintenance, and be provided complete with ladder access to AS 1657
- They shall be provided with a permanent cover that is designed to prevent the entry of dust, roof / ground water and prevent bird or animal life and shall be fitted with a close fitting & locked cover access opening.
- Venting shall be provided to ensure no excessive positive or negative pressures are transferred to the tank during inflow and outflow, in accordance with AS 2304.
- Air gaps, overflows & sludge points are provided in accordance with AS 3500
- Ensure correctly sized suction vortex plates are provided, complete with suction pipework & valving sized as required.

- Ensure all valves & access points are locked complete with 003 keyed locks to the satisfaction of the local authority.
- Ensure all pump suction valving is monitored to FIP to AS2118.1
- Ensure permanent signage & water level indicators are installed.
- Located: as noted, with final coordination and location to be determined on-site.

Allow to install tank in accordance with the tank manufacturers' requirements to suit the local conditions for structural, seismic, hydrostatic, wind & geo-technical support.

- Ensure to include into the O&M manual the required fire tank service requirements as noted within AS 2304. This is to include for monthly, yearly & ten-yearly service intervals as noted within AS 2304.

5.7 Testing

5.7.1 General

The installation shall be tested to the satisfaction of the Superintendent, the Fire Brigade and the relevant Statutory Authorities, prior to the acceptance of the installation and the commencement of the Defects Liability Period.

Provide all necessary skilled and competent personnel together with all the equipment and electric power required to test and commission the works.

Test the installation on completion to ensure that the installation complies with the specification and operates correctly under normal, emergency and fault conditions.

All testing and commissioning shall be pre-planned and scheduled in order to fully coordinate with other relevant trades and shall be carried out in an efficient manner with a minimum of inconvenience to all concerned.

All equipment or materials found to be faulty during testing shall be replaced free of charge.

Should a trial or test be deemed unsatisfactory by the Superintendent it shall be repeated at no further charge until satisfactory results are obtained.

5.7.2 Hydrostatic Testing

Flush the system with clean water.

All pipe reticulation shall be hydrostatic pressure tested for a minimum period of two hours at 1400kpa standing pressure.

In the event of any drop of pressure during the test periods, the leaks shall be found, made dry and the line retested.

The Superintendent shall be given 48 hours notice of intention to carry out testing.

5.7.3 Installation Completion Certificates

Submit prior to practical completion certificates as per Appendices A and B of the Code of Practice and AS2118 Part 10.

5.7.4 Water Quality Tests

5.7.4.1 Function

To demonstrate that the sprinkler system water is clean and free from foreign matter, including sand, sediments and other impurities.

5.7.4.2 Discharge test

Inspect water discharged from at least one of the flushing valves, for foreign matter.

5.7.4.3 Sprinkler head test

Remove at least 10% of the sprinkler heads, at designated locations, and inspect them for foreign matter lodged across the orifices of the heads.

5.7.4.4 Test failure

If foreign matter is observed in quantities likely to impair the functioning of the sprinkler system:

- Drain the system;
- Open and clean the alarm valve and strainer; and
- Recharge the system.

5.7.5 Regular Test

After the installation has been successfully acceptance tested, carry out monthly tests of the fire sprinkler installation, in accordance with the testing requirements of AS1851.

5.7.6 Annual Test

Complete an annual test of the complete system one month before the end of the maintenance period.

5.8 Records

Provide a good quality stiff plastic covered 100 page log book for record purposes, for the installation. Minimum page size shall be 114mm x 203mm.

Entries shall be made in the log book of all tests, alarms, maintenance, etc. as from the date of in-situ testing for commissioning of the system. All entries shall be made in ink.

5.9 Spares

The following spares shall be supplied as part of the contract:

Spare sprinkler heads as nominated in AS2118.

The spare sprinkler heads shall be provided within the metal residential sprinkler valve cabinet.

5.10 Completion

The whole of the installation shall be left complete in every way. After necessary tests required by the Fire Brigade have been carried out, the written approval of such Authority shall be handed to the Superintendent.

5.11 Training

Provide on site training for staff in the operation and testing of the system. Allow 2 sessions each of 1 hour duration.